

# NPR Locomotion: Improving Spatial Awareness and Cybersickness with Sketch-Like Continuous Transitions

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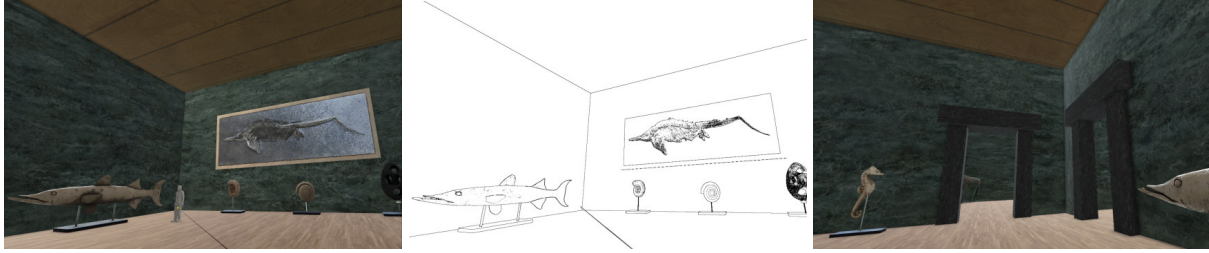


Figure 1: Left: Participant pointing to the avatar to go to the next location. Middle: During the transition in NPR condition. Right: Immediately after arriving at the new location and orientation.

**Abstract**—Architectural design reviews in Virtual Reality (VR) require clients to move in the environment to understand building layouts. However, locomotion methods present a trade-off: continuous movement supports spatial awareness but causes cybersickness, while teleportation prevents cybersickness but does not support spatial orientation as much. We investigated whether applying sketch-style non-photorealistic rendering (NPR) during continuous locomotion in guided architectural tours reduces cybersickness while maintaining the information necessary for spatial comprehension of the design. In a within-subjects study ( $N=36$ ), participants completed guided tours on three separate days under three different viewing conditions: full color continuous locomotion (FCL), NPR continuous locomotion, and jump teleportation (JMP). Each visit spanned a different set of rooms, with the correspondence between floorplan and viewing condition, as well as presentation order, counterbalanced between participants. We assessed cybersickness using pre-post SSQ surveys and the Fast Motion Sickness (FMS) scale, and spatial awareness using a point-to-unseen-landmark task, egocentric distance-to-landmark estimation, map drawing accuracy, and object recognition. Among the highlights of our findings are that pointing accuracy was significantly greater in the NPR than JMP condition, while Bayesian equivalence testing confirmed statistical equivalence between the NPR and FCL conditions. Additionally, post-exposure cybersickness was significantly lower in the NPR condition than FCL condition. Instantaneous discomfort scores were lowest in the JMP condition, intermediate with NPR, and highest with FCL. These findings demonstrate that deploying sketch-style rendering during guided transitions enables comfortable architectural exploration without sacrificing the spatial comprehension needed for design evaluation.

**Index Terms**—Virtual Reality, Guided Architectural Tours, Cybersickness, Non-Photorealistic Rendering, Spatial Cognition

## 1 INTRODUCTION

Architects increasingly use VR to present building designs to clients, as this enables immersive spatial experiences that are impossible with traditional drawings or physical models. A critical approach to demonstrating a design is the guided walkthrough, where the architect leads clients through the design by transitioning between waypoints that showcase the design. Ideally, VR locomotion techniques should enable the user to travel in the model while maintaining both spatial awareness and physical comfort. However, current VR locomotion methods introduce a trade-off. When users glide through a virtual space, they get the continuous visual feedback that is essential for understanding spatial relationships and building layouts [4, 30]. Clients can intuitively observe how rooms connect, easily appreciate distances between spaces, experience the circulation flow through the building [9, 31] and develop a coherent mental model of the design. However, continuous movement introduces sensory conflict between visual motion and stationary vestibular signals that can lead to cybersickness [32, 35]. Cybersickness symptoms include but are not limited to nausea, disorientation, and visual discomfort [10, 11, 38]. Cybersickness limits the duration of exposure that users can comfortably tolerate, and can force early session termination [41, 42], which is problematic when extensive consideration is required. Concerns about cybersickness pose a particular obstacle to using VR to review architectural designs in contexts where

users must explore an entire building layout, which can mean traversing hundreds of meters across multiple floors and rooms. Teleportation addresses cybersickness by eliminating visual motion, so that clients instantly “jump” to new locations without traversing the path between origin and destination. This is effective at preventing cybersickness [2], but eliminates the spatial continuity that helps clients keep track of where they are in relation to where they have been [4, 7, 30]. This is particularly problematic in the context of guided architectural walkthroughs, in which the transitions between waypoints typically involve a change in viewing direction as well as a change in location. This paper investigates a solution inspired by architectural design procedures: non-photorealistic rendering (NPR). Architects use sketch-style drawings in the early stages of the design process to help clients focus on the structure of the space and avoid premature fixation on surface details [39]. We anticipated that applying similar sketch-style rendering during locomotion transitions will reduce cybersickness by removing high-frequency textural details while at the same time supporting spatial awareness by preserving the essential structural content necessary for architectural comprehension.

## 2 RELATED WORK

### 2.1 Locomotion for Architectural Guided Tours

Architectural VR applications require methods for transitioning clients to demonstrate a design. Unlike gaming or training applications where users freely explore environments, architectural presentations utilize guided tours to ensure clients experience spaces in intended sequences and orientations. Locomotion techniques can be grouped by the degree

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of freedom that the user has. In active locomotion, users can control their movement using a joystick, gestures, physical walking, or other interactive means. Active control can reduce cybersickness [1, 18], but it introduces problems for architectural presentations: clients may deviate from intended paths, miss critical viewpoints, or become distracted by the navigation mechanism rather than focusing on design content. In passive locomotion, movement occurs without the user having control over it. This better suits the architectural design review context as it ensures clients experience the presented spaces in the intended way. Additionally, locomotion can be continuous or discrete. In continuous locomotion, the user travels smoothly through the virtual space, receiving continuous visual feedback about their surroundings. Research has shown that continuous transitional visual feedback can better support spatial awareness relative to discrete relocation, enabling more accurate model comprehension [4, 30]. Continuous passive locomotion, sometimes called "dragging", often triggers cybersickness because users experience visual motion without corresponding vestibular or proprioceptive signals [10, 36]. On the other hand, during discrete locomotion (e.g. teleportation, jumping) users are instantly relocated to new locations without intermediate visual feedback. This is effective at mitigating cybersickness by minimizing visual motion [2, 5], but the client loses the spatial continuity. Studies have shown that teleportation can adversely affect spatial memory, including the direction and distance to previously-visited landmarks, and that it impedes coherent mental map formation [4, 7, 30]. For architectural guided tours, continuous passive locomotion is the ideal approach, as it maximizes the ability of the client to understand and appreciate the space while enabling the designer to curate their experience by maintaining control over where they go and what they see. However, the combination of passive control and continuous visual motion creates optimal conditions for nausea and discomfort [11, 42].

## 2.2 Approaches to Cybersickness Mitigation

### 2.2.1 Machine Learning Approaches

Machine Learning (ML) methods have been explored to enable the more objective, automated and real-time detection and prediction of cybersickness [47, 48]. Early methods leveraged physiological data like heart rate and galvanic skin response [12–14]. More recent approaches combine physiological data with head-tracking and eye-tracking [21, 22, 34]. Enhanced prediction architectures like MAC (Multimodal Attention-Based Cybersickness Prediction) fuse multiple data streams for an early warning system [23]. Relying on the detection of early physiological indicators of cybersickness can be useful for forestalling the worsening of symptoms, but doesn't prevent their onset. Many ML approaches therefore aim to forecast cybersickness onset 30–60 seconds before it occurs [21, 22]. The Cybersickness Detection and Mitigation System (CDMS) [44] uses electroencephalogram (EEG) signals to dynamically predict cybersickness onset and then modifies VR parameters like field of view and motion speed to mitigate symptoms, however adjustments like these may limit spatial understanding [2, 47] and disrupt the natural flow of an architectural presentation. Additionally, methods that require specialized sensor equipment (EEG, etc.) are impractical for client presentations. NPR stylization offers an alternative strategy for preventing cybersickness rather than detecting and reacting to it.

### 2.2.2 NPR in Architectural Contexts

NPR rendering of virtual environments has been demonstrated at interactive rates [27], and distance perception in immersive line-drawing environments has been found comparable to that in photorealistically rendered ones [17]. Transitioning to an NPR stylization during locomotion could mitigate cybersickness by reducing high-frequency textural details that exacerbate the experience of incongruent optical flow [13, 24, 40]. By preserving structural edges corresponding to walls, doorways, and major architectural features, NPR stylization retains the geometric information necessary for spatial understanding. Strothotte et al. [43] used line-drawing to visualize virtual reconstructions of ancient architecture, exploiting the ability of sketch-like rendering to communicate structural form while signaling the uncertainty of unverified details. Schumann et al. [39] surveyed architects and found that

sketch-style renderings of CAD models were judged more appropriate than photorealistic shaded renderings for presenting preliminary designs as they were more effective at stimulating discussion and affording the possibility for change. Using a video-see-through HMD, Vaziri et al. [46] found that egocentric distance estimation accuracy in real-world building interiors was unaffected when the live camera images were converted to line-drawings. Recently, Vaziri [45] proposed using line-drawing-style transitions in guided architectural tours, but no previous work has yet empirically evaluated whether such an approach will work to successfully reduce cybersickness and at the same time preserve the spatial awareness necessary for architectural design comprehension.

## 2.3 Research Questions and Hypotheses

This paper addresses three questions for architectural VR applications:

1. RQ1: Can transitioning to an NPR style during continuous locomotion between waypoints reduce cybersickness compared to locomotion in a full detail (FCL) environment?
2. RQ2: Can spatial awareness be maintained more effectively by continuously moving in an line-drawing-style environment than by using discrete jumping teleportation?
3. RQ3: Can we achieve the "best of both worlds" by strategically deploying NPR stylization during transitions between waypoints on a guided architectural tour?

Our hypotheses follow from two theoretical premises that jointly define the comfort–cognition trade-off. First, cybersickness during passive locomotion is driven largely by the magnitude of optical flow generated by transitional visual motion [13, 40]: FCL presents full-detail visual motion, NPR attenuates the high-frequency component of that motion while retaining structural edges, and JMP eliminates transitional visual motion entirely. Second, spatial awareness benefits from the availability of continuous transitional visual feedback [4, 30]: both continuous methods (FCL and NPR) provide this, whereas JMP does not. Accordingly, we hypothesize that:

1. H1: Cybersickness will increase with the amount of transitional visual motion presented: NPR locomotion will cause less cybersickness than FCL, and JMP will cause the least.
2. H2: Both continuous methods will provide better spatial awareness than Jump teleportation; in particular, NPR locomotion will better support spatial awareness than JMP.
3. H3: Because the structural edges preserved by NPR stylization carry the visual information needed for spatial updating, NPR locomotion will support spatial awareness to a similar extent as FCL.

Confirming these hypotheses would support the notion that transiently deploying NPR stylization offers a 'win-win' solution for supporting both comfort and cognition during architectural guided tours. We return to each hypothesis explicitly in Section 5.

## 3 METHOD

### 3.1 Design

We designed a within-subjects experiment to compare the three VR locomotion methods across three similar but distinct virtual museum floor plans, which participants experienced in counter-balanced order on different days. The environments were built in Unity 3D 2022.3.10f1 LTS using museum artifacts that were purchased from the Unity Store, and were all rendered at 120 FPS.

#### 3.1.1 Hardware Setup

The virtual environments were experienced using an HTC Vive Pro 2, with dual 2448×2448 pixel displays, a 117°H × 96°V field of view, a 120Hz refresh rate, and SteamVR BaseStation 2.0 tracking.

### 3.1.2 NPR Rendering Pipeline

The NPR effect is produced by a custom post-processing full-screen outline shader. It is implemented in Unity’s Universal Render Pipeline (URP) Shader Graph. The idea comes from a publicly available outline shader [20] that detects edges from colors and normals over a non-weighted  $3 \times 3$  kernel. However, we combined depth-based and normal-based edge detection over a Gaussian-weighted  $5 \times 5$  kernel, which produces smoother and more consistent edges.

For normal-based edges, three concurrent pipelines sample the world-space normal buffer (URP "Sample Buffer" node, *SB*, the Source Buffer set to "Normal World Space"), and the dot products (angles) between normals at opposing offsets around each fragment ( $i, j$ ) are combined with Gaussian weights:

$$\begin{aligned} normal(i, j) = & 0.5 \times [SB(i+1, j) \cdot SB(i-1, j) + SB(i, j+1) \cdot SB(i, j-1)] \\ & + 0.2 \times [SB(i+2, j) \cdot SB(i-2, j) + SB(i, j+2) \cdot SB(i, j-2)] \\ & + 0.3 \times [SB(i+1, j+1) \cdot SB(i-1, j-1) \\ & + SB(i-1, j+1) \cdot SB(i+1, j-1)] \end{aligned} \quad (1)$$

Depth-based edges use three analogous pipelines that sample scene depth (URP "Scene Depth" node, *SD*, with "Eye" sampling mode) and we use subtraction (depth difference) in place of the dot product, combined with the same Gaussian weights.

Dense texture detail on distant floors and ceilings produces false-positive edges that render as solid black regions near the horizon, where these surfaces are viewed at grazing angles. To suppress these, the floor and ceiling objects are hidden during the edge computation step, and the normal-edge response is gated by the viewing angle: we take the absolute dot product between the world-space surface normal  $N$  and the view direction  $V$ , which approaches zero as a surface is viewed edge-on, and threshold it with a *ViewDirectionThreshold* so that edges are retained only where the surface faces the viewer:

$$\begin{aligned} wnormals(i, j) = & [1 - \text{STEP}(normal(i, j), NormalThreshold)] \\ & \times \text{STEP}(|N \cdot V|, ViewDirectionThreshold) \end{aligned} \quad (2)$$

$$edge(i, j) = wnormals(i, j) + \text{STEP}(depth(i, j), DepthThreshold) \quad (3)$$

The final edge value is saturated, multiplied by the outline color, and inverted, yielding dark structural lines on a white background (Figure 1, middle). A *NormalThreshold* of 1.75, a *DepthThreshold* of 0.3, and a *ViewDirectionThreshold* of 0.3 produced the best results across our environments. The NPR view is active during the entire transition, i.e., during both the translation toward the avatar’s position and the subsequent rotation into its facing direction, and the view is transitioned back to full detail upon arrival. Benchmarking confirmed that the frame rate remained at 120 FPS in both the NPR and FCL conditions.

### 3.1.3 Participants and Procedure

We recruited 36 participants to complete three sessions, with at least 48 hours between the consecutive sessions to minimize carry-over effects. Each session involved one of the three locomotion techniques and one of the virtual museums.

1. Full Color Detail (FCL): Continuous locomotion at 1.5 m/s with full color rendering and texture detail.
2. Non-Photorealistic Rendering (NPR): Continuous locomotion at 1.5 m/s with sketch-style rendering during transitions.
3. Jump (JMP): Instant teleportation, which provides no visual feedback during transitions.

Note that the three conditions vary along two design dimensions: rendering style during the transition (full-color textured vs. sketch-style NPR) and transition type (continuous vs. discrete). The labels FCL, NPR, and JMP emphasize the most diagnostic feature of each condition: FCL and NPR share continuous transitions but differ in rendering style, while JMP differs from both in transition type. Participants were asked

to explore the virtual museum through a guided tour, in which they traveled between pre-defined waypoints. When it was time to move to the next waypoint, the new location and facing direction would be transiently indicated by a humanoid avatar. Participants activated the transition by pointing to the avatar using a ray cast from a controller in their dominant hand. After they confirmed the selection by clicking, the avatar disappeared and they moved into alignment with it. In the continuous movement conditions they first translated to the avatar’s position and then rotated into alignment with the direction the avatar was facing. After the transition, participants were asked to point to their previous three stopping locations and verbally estimate their distance from each one using the measurement unit of their choice. The paths through each museum were designed so that at each waypoint only their immediately previous position was visible and the second and third points back were occluded by walls. This required participants use their memory of where they had come from, rather than relying on line-of-sight landmarks. Each tour traversed the museum three times, visiting a total of 32 different waypoints. The floorplans and paths are shown in figure 2. The presentation order of the three locomotion methods, and the association of locomotion method with floor plan, were counterbalanced between participants to control for order effects.

| P  | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | P  | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | P  | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> |
|----|-----------------|-----------------|-----------------|----|-----------------|-----------------|-----------------|----|-----------------|-----------------|-----------------|
| 1  | F1              | F2              | F3              | 13 | F2              | F1              | F3              | 25 | F3              | F1              | F2              |
|    | FCL             | JMP             | NPR             |    | FCL             | NPR             | JMP             |    | FCL             | JMP             | NPR             |
| 2  | F1              | F3              | F2              | 14 | F2              | F3              | F1              | 26 | F3              | F2              | F1              |
|    | FCL             | NPR             | JMP             |    | FCL             | JMP             | NPR             |    | FCL             | NPR             | JMP             |
| 3  | F1              | F2              | F3              | 15 | F2              | F3              | F1              | 27 | F3              | F2              | F1              |
|    | FCL             | NPR             | JMP             |    | FCL             | NPR             | JMP             |    | FCL             | JMP             | NPR             |
| 4  | F1              | F3              | F2              | 16 | F2              | F1              | F3              | 28 | F3              | F1              | F2              |
|    | FCL             | JMP             | NPR             |    | FCL             | JMP             | NPR             |    | FCL             | NPR             | JMP             |
| 5  | F1              | F2              | F3              | 17 | F2              | F1              | F3              | 29 | F3              | F1              | F2              |
|    | NPR             | FCL             | JMP             |    | NPR             | JMP             | FCL             |    | NPR             | FCL             | JMP             |
| 6  | F1              | F3              | F2              | 18 | F2              | F3              | F1              | 30 | F3              | F2              | F1              |
|    | NPR             | JMP             | FCL             |    | NPR             | FCL             | JMP             |    | NPR             | JMP             | FCL             |
| 7  | F1              | F3              | F2              | 19 | F2              | F3              | F1              | 31 | F3              | F2              | F1              |
|    | NPR             | FCL             | JMP             |    | NPR             | JMP             | FCL             |    | NPR             | FCL             | JMP             |
| 8  | F1              | F2              | F3              | 20 | F2              | F1              | F3              | 32 | F3              | F1              | F2              |
|    | NPR             | JMP             | FCL             |    | NPR             | FCL             | JMP             |    | NPR             | JMP             | FCL             |
| 9  | F1              | F2              | F3              | 21 | F2              | F1              | F3              | 33 | F3              | F1              | F2              |
|    | JMP             | NPR             | FCL             |    | JMP             | FCL             | NPR             |    | JMP             | NPR             | FCL             |
| 10 | F1              | F3              | F2              | 22 | F2              | F3              | F1              | 34 | F3              | F2              | F1              |
|    | JMP             | FCL             | NPR             |    | JMP             | NPR             | FCL             |    | JMP             | FCL             | NPR             |
| 11 | F1              | F3              | F2              | 23 | F2              | F3              | F1              | 35 | F3              | F2              | F1              |
|    | JMP             | NPR             | FCL             |    | JMP             | FCL             | NPR             |    | JMP             | NPR             | FCL             |
| 12 | F1              | F2              | F3              | 24 | F2              | F1              | F3              | 36 | F3              | F1              | F2              |
|    | JMP             | FCL             | NPR             |    | JMP             | NPR             | FCL             |    | JMP             | FCL             | NPR             |

Table 1: Counterbalanced assignment of floor plan (F1–F3) and locomotion method to each of the three sessions (1<sup>st</sup>–3<sup>rd</sup>) for the 36 participants (P). FCL = Full Color, JMP = Jump, NPR = Non-Photorealistic Rendering.

## 3.2 Virtual Environments

We chose virtual museums as the experimental environment primarily for experimental control rather than as a claim that museums represent typical architectural designs. Our goal was to realize three environments that were qualitatively similar but very different in their particulars, which included having an arbitrary, non-standard layout with rooms whose sizes, shapes, and contents could not be readily predicted from the sizes, shapes, and contents of other rooms in the same building. The essential architectural elements (walls, doorways, room boundaries, and room-to-room transitions) are common to other architectural settings, and we used high-frequency textures to compensate in part for the paucity of higher-frequency geometric details relative to production-quality architectural models.

The three floor plans share identical overall dimensions (80m  $\times$  50m), identical room counts (six halls each), and identical path structure (32 waypoints across three consecutive end-to-end traversals), while differing in architectural details and artifacts. The qualitative similarity

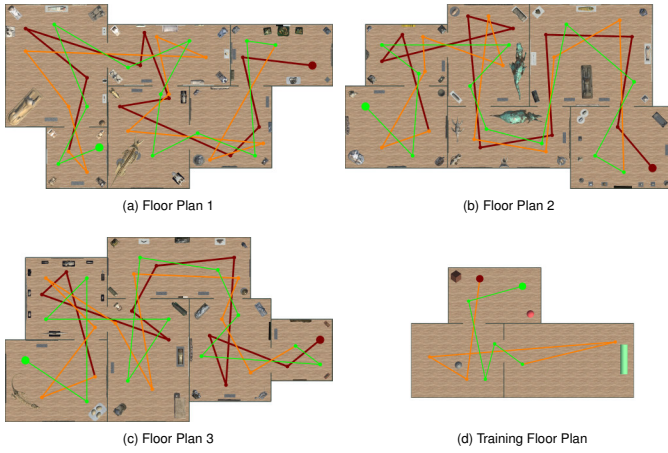


Figure 2: Path segments for the three floor plans (a-c) and the training floor plan (d). Green: starting path segment; orange: returning path segment; red: final path segment.

in the design of the waypoint layouts normalizes the pointing task demands across floor plans. Artifacts (Egyptian, Buddhist, Roman, and fossil collections) were curated so that each floorplan contained thematically similar but visually distinct items, to minimize cross-environment interference in the object recognition task, and similar items were grouped together within each floorplan to provide each room with a distinct identity. Consistent with this balancing, the post-hoc analyses reported in Section 4 found no significant floor-plan effect on any of the dependent measures tested.

### 3.3 Outcome Measures

We used the following measures to assess spatial awareness, cybersickness, and user experience:

- **Baseline measures:** spatial orientation ability (SOT [15]) and motion sickness susceptibility (VIMSSQ short form [28]). These were collected once, in Session 1, to characterize individual differences and serve as covariates in the analyses.
- **Directions to previously visited waypoints:** At each station, participants used the controller to point to the three most recent prior waypoints. Throughout the paper, we refer to these targets by their  $n$ -back level: the 1-back target is the immediately previous waypoint, the 2-back target the one before it, and the 3-back target the one before that. This task assesses spatial updating, i.e., participants' continuous maintenance of self-position and orientation during travel. Pointing to previously visited locations has been shown to be a sensitive measure of spatial updating during locomotion [26, 37]. We recorded the unsigned angular error between the pointing direction and the true direction to the target.
- **Distance to previously visited waypoints:** After indicating the direction to each previously-visited location, participants provided a verbal estimate of their egocentric distance from that target location, using the measurement unit of their choice. This measure probes egocentric distance representations separately from the directional knowledge assessed by pointing; research on distance perception has examined the visual cues underlying such egocentric distance judgments [19, 33]. We recorded the signed distance error to reflect the direction of estimation bias.
- **Spatial memory:** After each session, participants were asked to draw a floorplan-style sketch map of the environment including the landmarks visited, which we scored on correct room count, door placement, and artifact descriptions [16, 29]. Sketch maps assess the survey-level mental representation of the layout formed during the tour. Participants also completed an object recognition test (yes/no judgments for 12 artifact images: 6 present in their

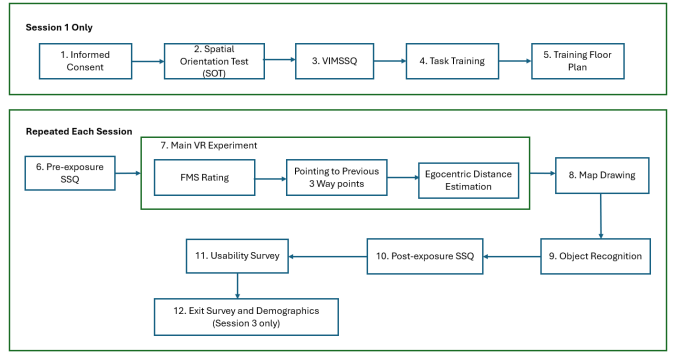


Figure 3: Experiment procedure. The top region shows steps completed in Session 1 only. The bottom region shows steps repeated in each of the three sessions.

environment, 6 absent), which assesses incidental encoding of environment content.

- **Cybersickness:** We used two complementary instruments. At the beginning and end of each session, we administered the SSQ [8] and calculated the change (post-exposure — pre-exposure) in total score and in the nausea, oculomotor and disorientation subscales;  $\Delta$ SSQ captures the accumulated symptom increase produced by the full exposure. Additionally, at each waypoint, after making their spatial judgments, participants verbally reported their current level of discomfort on the Fast Motion Sickness scale (FMS, 0–10) [25]; these repeated instantaneous self-reports capture the progression of cybersickness over the course of the session.
- **User experience:** per-session ratings of preference, presence, naturalness, and realism on 4-point scales, and a summative exit survey comparing all three methods with open-ended feedback (Session 3 only).

### 3.4 Experiment's Procedure

The study consisted of three sessions separated by at least 48 hours. The first session took approximately 45 minutes as it included an introduction and guide to the study but the two subsequent sessions took approximately 30 minutes. Figure 3 illustrates the complete procedure. Session 1 began with five preparatory steps: (1) obtaining informed consent approved by our Institutional Review Board (IRB), (2) administering the Spatial Orientation Test (SOT) [15] as a baseline measure of spatial ability, (3) completing the VIMSSQ short form [28] to assess motion sickness susceptibility, (4) task instructions, and (5) completing training in a simplified virtual environment containing three rooms and nine waypoints. The following steps are for each session: (6) pre-exposure SSQ [8], (7) traverse the virtual environment, traveling from waypoint to waypoint within each floor plan using one locomotion method, completing spatial judgment tasks at each waypoint as described earlier and verbally reporting discomfort at each station on the Fast Motion Sickness Scale (FMS, 0–10) [25], (8) draw a floorplan-style map of the virtual environment, including the landmarks visited [16, 29], (9) object recognition survey, answering multiple yes/no questions about whether objects (shown in pictures) had been seen while in the museum, (10) post-exposure SSQ, and (11) usability survey rating preference, presence, naturalness, and realism on 4-point scales. In the third session only, participants also completed (12) an exit survey collecting overall method preference and demographic information.

### 3.5 Participants

We recruited 36 participants through flyers posted on the University of Minnesota campus, announcements on student housing social media, and in-class announcements. Participant age ranged from 18 to 49 (Mean = 24.53, Median = 23, Standard Deviation = 6.29). Participants were compensated with \$45 for completing all three sessions. Additionally, to incentivize diligence, we awarded a \$10 bonus to the top 25%

of participants who most accurately completed the spatial judgment tasks. Participants completed IRB-approved informed consent forms. Inclusion criteria were: age 18+ years, normal or corrected-to-normal vision, not pregnant, and no history of epilepsy, seizures, or severe postural instability. Nineteen participants identified as men and 17 as women. Participants represented diverse ethnic backgrounds: Western Europe ( $n=10$ , 28%), Eastern Europe ( $n=4$ , 11%), North America ( $n=1$ , 3%), Central American and Caribbean ( $n=1$ , 3%), North Africa ( $n=1$ , 3%), Sub-Saharan Africa ( $n=1$ , 3%), West Asia / Middle East ( $n=4$ , 11%), South and Southeast Asia ( $n=12$ , 33%), and East and Central Asia ( $n=7$ , 19%). Regarding race, participants self-identified as Asian (Indian, Chinese, Japanese, Korean, Singaporean) ( $n=19$ , 53%), Middle Eastern or North African ( $n=5$ , 14%), Black ( $n=1$ , 3%), and White ( $n=12$ , 33%). Prior HMD experience varied: 7 participants (19%) had no experience, 19 (53%) had used HMDs rarely (once or twice total), 5 (14%) used them occasionally (2-3 times per month), and 5 (14%) used them often (weekly or more).

## 4 RESULTS

### 4.1 Cybersickness

#### 4.1.1 FMS

Shapiro-Wilk tests indicated that FMS distribution is not normal, so we ran a Friedman test, which showed a significant effect of locomotion method on FMS scores ( $\chi^2 = 10.27$ ,  $p = 0.006$ ,  $W = 0.14$ ). Mean FMS scores had a consistent pattern across conditions: FCL ( $M = 1.48$ ,  $SD = 1.43$ ) > NPR ( $M = 1.19$ ,  $SD = 1.56$ ) > JMP ( $M = 0.88$ ,  $SD = 1.53$ ). Post-hoc Wilcoxon signed-rank tests (Bonferroni-corrected) showed no statistically significant pairwise differences: FCL vs. JMP ( $p = 0.071$ ), FCL vs. NPR ( $p = 0.399$ ), and NPR vs. JMP ( $p = 0.175$ ). However, we noticed that during the first 10 stations all three conditions produced similarly low FMS levels with no significant pairwise differences (all  $p > 0.13$ ; mean differences  $< 0.16$ ). From station 11 onward, FCL produced significantly higher FMS than JMP ( $W = 93.5$ ,  $p = 0.007$ , mean difference = 1.08) and the NPR vs. JMP difference was also significant ( $W = 74.0$ ,  $p = 0.017$ , mean difference = 0.60) under Holm-Bonferroni correction. FCL vs. NPR did not reach significance ( $p = 0.069$ ). To further characterize how the conditions diverge over time, we also compared FMS scores between methods at each station ( $n = 32$ –36 per station; Figure 4). Per-station Friedman tests were not significant at any of the first 16 stations (all  $p > 0.05$ ; stations 1–10: all  $p > 0.17$ ), but reached significance at 15 of the final 16 stations (stations 17–32, uncorrected  $p < 0.05$  at all but station 26; 11 of these stations, including all of 18–23 and 28–32, remained significant under Benjamini-Hochberg FDR correction). Exploratory pairwise comparisons trace the order in which the conditions separate: FCL produced higher FMS than JMP beginning at station 11 (uncorrected  $p < 0.05$  at 18 of the remaining 22 stations), whereas FCL did not begin to differ from NPR until station 21 ( $p < 0.05$  at 9 of the remaining 12 stations), and NPR and JMP differed only sporadically (4 stations). Cybersickness thus accumulated fastest under FCL, at an intermediate rate under NPR, and slowest under JMP: all three conditions start near zero, the FCL-JMP separation emerges first, and the FCL-NPR separation appears only after sustained exposure, consistent with the split analysis above.

#### 4.1.2 SSQ

A Friedman test on SSQ change scores (post-exposure – pre-exposure) showed a significant effect of locomotion method on Total severity ( $\chi^2 = 7.49$ ,  $p = 0.024$ ). Mean  $\Delta$ SSQ Total scores were FCL ( $M = 28.57$ ), NPR ( $M = 21.61$ ), and JMP ( $M = 15.58$ ).

Post-hoc Wilcoxon signed-rank tests with Holm-Bonferroni correction revealed that FCL caused significantly higher Total symptoms than both JMP ( $p = 0.009$ , adjusted  $\alpha = 0.017$ ) and NPR ( $p = 0.018$ , adjusted  $\alpha = 0.025$ ). JMP and NPR did not differ significantly ( $p = 0.120$ ). The effect was most pronounced for Nausea ( $\chi^2 = 14.28$ ,  $p < 0.001$ ), where FCL produced significantly higher increases than JMP ( $p = 0.005$ ) and NPR ( $p = 0.003$ ). Oculomotor ( $\chi^2 = 4.08$ ,

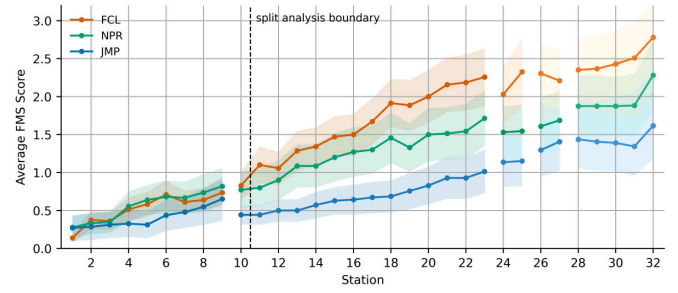


Figure 4: Mean FMS ( $\pm 1$  SEM) at each station by locomotion method. The dashed line marks the boundary used in the split analysis (stations 1-10 vs. 11+). The gaps in the lines are where participant drop-outs required the average FMS scores to be computed over a fewer number of people.

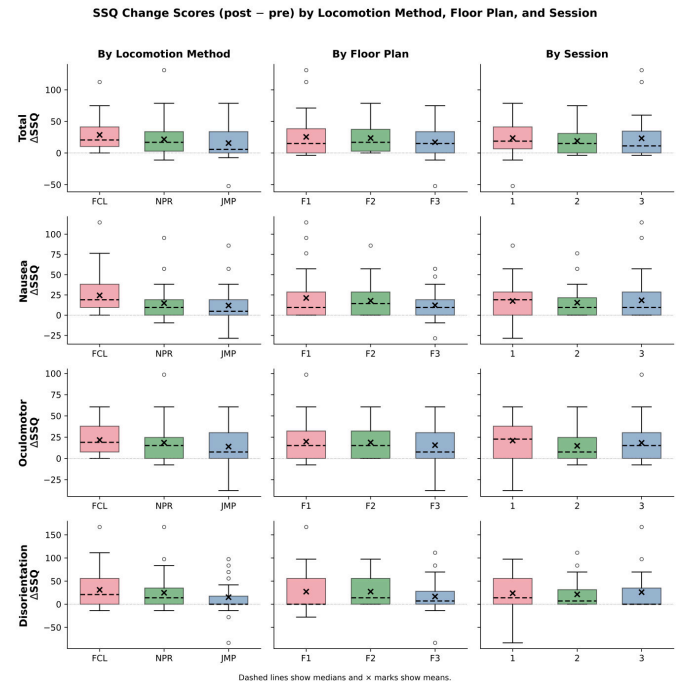


Figure 5: SSQ change scores (post – pre) by locomotion method, floor plan, and session, for the Total score and the Nausea, Oculomotor, and Disorientation subscales (all using standard weighted SSQ scoring). In each box the dashed line is the median and the black  $\times$  is the mean; open circles are outliers beyond  $1.5 \times IQR$ , and the dotted line marks zero change.

$p = 0.130$ ) and Disorientation ( $\chi^2 = 8.71$ ,  $p = 0.013$ ) subscales followed a similar pattern; for Disorientation, FCL was significantly higher than JMP ( $p = 0.021$ ) but not NPR ( $p = 0.089$ ). Friedman tests on  $\Delta$ SSQ by session revealed no significant adaptation effect for Total ( $\chi^2 = 1.72$ ,  $p = 0.423$ ), Nausea ( $\chi^2 = 1.04$ ,  $p = 0.595$ ), Oculomotor ( $\chi^2 = 2.20$ ,  $p = 0.333$ ), or Disorientation ( $\chi^2 = 0.57$ ,  $p = 0.752$ ). Floor plans had no significant effect on any  $\Delta$ SSQ measure (Total:  $p = 0.718$ ; Nausea:  $p = 0.847$ ; Oculomotor:  $p = 0.906$ ; Disorientation:  $p = 0.145$ ), confirming consistent design across the three virtual environments. Figure 5 shows  $\Delta$ SSQ scores by session.

### 4.2 Spatial Awareness

#### 4.2.1 Pointing Data

We identified two participants (P5, P8) with mean absolute errors exceeding three standard deviations above the group mean, Figure 6, and these were therefore excluded from this analysis. We analyzed

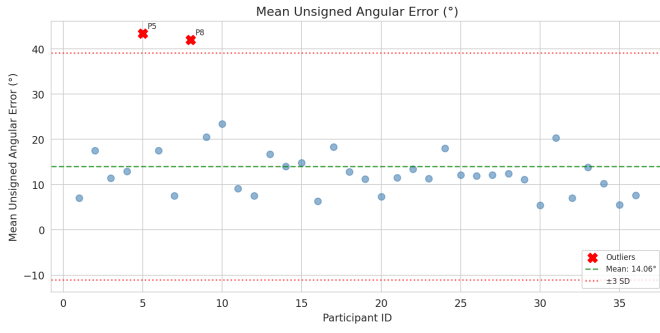


Figure 6: Outlier Detection for Pointing Data

unsigned angular error for pointing to the three previous waypoints across all three conditions. Shapiro-Wilk tests indicated non-normal distributions for the NPR condition ( $W = 0.91$ ,  $p = 0.009$ ), so we used Friedman tests for analysis. A Friedman test showed a significant effect of locomotion condition on unsigned angular error ( $N = 34$ ,  $\chi^2 = 15.76$ ,  $p < 0.001$ ), indicating that locomotion method influenced spatial orientation performance. Mean unsigned errors were  $10.60^\circ$  for FCL (rank = 1.59),  $11.59^\circ$  for NPR (rank = 1.88), and  $15.10^\circ$  for JMP (rank = 2.53). Post-hoc Wilcoxon signed-rank tests with Holm-Bonferroni correction showed that FCL and NPR did not differ significantly ( $p = 0.143$ ), while both continuous locomotion methods produced significantly lower pointing errors than JMP (FCL vs JMP:  $p < 0.001$ ; NPR vs JMP:  $p < 0.001$ ). To statistically show the equivalence in H3, we ran a Bayesian equivalence analysis that compares NPR and FCL pointing errors. For each participant, we computed the mean difference in unsigned angular error between conditions (NPR – FCL:  $M = 0.99^\circ$ ,  $SD = 3.43^\circ$ , Cohen’s  $d = 0.29$ ). Because no consensus equivalence margin has been established for pointing-based spatial orientation measures, we evaluated the Bayes Factor across a range of equivalence margins ( $\pm 3^\circ$  to  $\pm 15^\circ$ ) and Cauchy prior scales ( $r = 0.5$  to  $r = 2.0$ ):  $BF_{01}$  consistently exceeded 1000 in all configurations, strongly supporting equivalence over difference. By Jeffreys’ convention, this strong evidence supports that NPR locomotion preserves spatial orientation accuracy comparable to FCL continuous movement. This result supports our hypothesis that NPR locomotion preserves spatial awareness comparable to FCL continuous movement, while jump teleportation that eliminates transitional visual information impedes spatial updating. Floor plans had no significant effect on unsigned angular error ( $\chi^2 = 1.12$ ,  $p = 0.572$ ), confirming consistent spatial layout design across the three virtual environments.

To examine the relationship between spatial orientation ability and pointing performance, we correlated each participant’s SOT error (mean angular error across 12 trials, where higher SOT error indicates lower spatial ability) with their mean unsigned pointing error averaged across all three locomotion conditions including all 36 participants. P5 and P8 were excluded from the pointing analysis as outliers, but their data were included to examine the spatial ability and performance in VR. Therefore, correlation included all 36 participants. However, we excluded from the SOT calculation trials where no response was recorded within the given time limit, where the software had filled the error with  $90^\circ$  by default. Eleven participants ran out of time in at least one of the trials. The overall correlation was significant ( $r(34) = 0.580$ ,  $p < .001$ ), and after excluding the unfinished trials before computing the mean SOT for each participant, the correlation strengthened ( $r(34) = 0.699$ ,  $p < .001$ ). This confirms that higher SOT error (lower spatial ability) predicted larger mean unsigned pointing error. The correlations between the pointing error and SOT error per condition were also significant: FCL ( $r = 0.526$ ,  $p < 0.001$ ), NPR ( $r = 0.638$ ,  $p < 0.001$ ), and JMP ( $r = 0.613$ ,  $p < 0.001$ ). To study the locomotion method’s effect in this relation, we ran an ANCOVA across all 36 participants with locomotion method as a factor and corrected SOT score as a continuous covariate. The SOT covariate was a highly significant predictor of pointing error ( $F(1, 102) = 53.98$ ,  $p < 0.001$ ). The main effect of condition

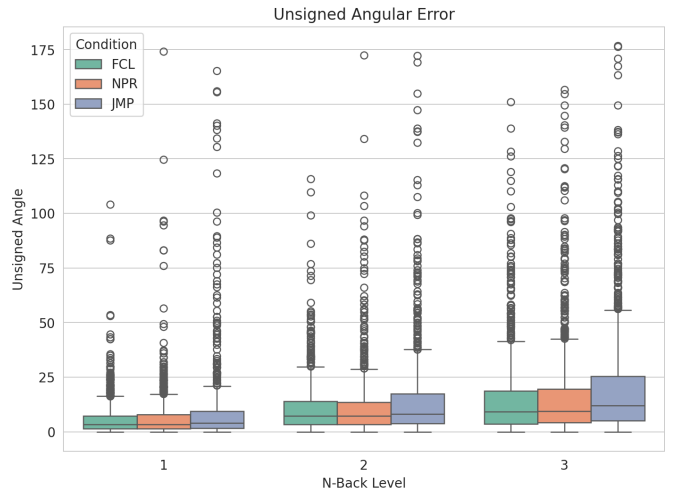


Figure 7: Angular Error After Removing Outliers

remained significant ( $F(2, 102) = 3.43$ ,  $p = 0.036$ ), and the Condition  $\times$  SOT interaction was also significant ( $F(2, 102) = 5.17$ ,  $p = 0.007$ ). This indicates that the relationship between spatial ability and pointing accuracy changed in magnitude across locomotion methods.

VIMSSQ scores showed a correlation with average  $\Delta$ SSQ across all three conditions that was marginal by Pearson ( $r(34) = .316$ ,  $p = .060$ ) and significant by Spearman ( $\rho(34) = .362$ ,  $p = .030$ ). The relationship between VIMSSQ and cybersickness remained notable across locomotion methods when examined by condition. The strongest relation appeared in JMP, where participants who reported greater susceptibility experienced correspondingly higher symptom increases ( $r(34) = .363$ ,  $p = .030$ ). FCL showed a similar but weaker trend ( $r(34) = .314$ ,  $p = .062$ ). However, NPR showed no relationship between susceptibility and cybersickness ( $r(34) = .082$ ,  $p = .636$ ). This pattern was confirmed in the absolute post-exposure SSQ scores: VIMSSQ predicted post-SSQ severity under FCL ( $r(34) = .379$ ,  $p = .023$ ) and JMP ( $p(34) = .474$ ,  $p = .004$ ), but not under NPR ( $r(34) = .069$ ,  $p = .691$ ). Across sessions, the VIMSSQ and  $\Delta$ SSQ correlation was most apparent in Session 1 ( $r(34) = .304$ ,  $p = .072$ ) and diminished in later sessions, confirming a general adaptation and learning effect. That susceptibility did not predict cybersickness in the NPR condition suggests that sketch-style rendering may decouple individual vulnerability from symptom outcomes, likely by reducing the high-frequency visual detail that affects susceptible individuals.

#### 4.2.2 Distance Estimates

We used signed distance error as the metric to reflect the direction of estimation bias, where negative values indicate underestimation and positive values indicate overestimation. Participants estimated egocentric distances to each pointed location using their preferred measurement units. As shown in Figure 8, two participants (P5, P34) who had more than 3 SD error were excluded from the analysis. Friedman tests revealed no significant effect of locomotion method on absolute signed error ( $\chi^2 = 1.65$ ,  $p = 0.439$ ). All three conditions produced similar underestimation biases (FCL:  $M = -10.50m$ ; NPR:  $M = -10.70m$ ; JMP:  $M = -10.76m$ ) which suggests that distance judgments were unaffected by the locomotion method. Floor plans also showed no significant effect on signed error ( $\chi^2 = 2.53$ ,  $p = 0.282$ ), with similar underestimation across F1 ( $M = -10.25m$ ), F2 ( $M = -10.92m$ ), and F3 ( $M = -10.80m$ ), which confirms that the three environments were comparable in their design characteristics. However, n-back level significantly influenced signed distance error ( $\chi^2 = 26.12$ ,  $p < 0.001$ ). Underestimation increased as the n in n-back increased: 1-back had a mean error of  $M = -9.24m$ , 2-back  $M = -11.12m$ , and 3-back  $M = -11.69m$ . All pairwise comparisons were significant after Holm-Bonferroni correction (all  $p < 0.003$ ). Session order had no significant

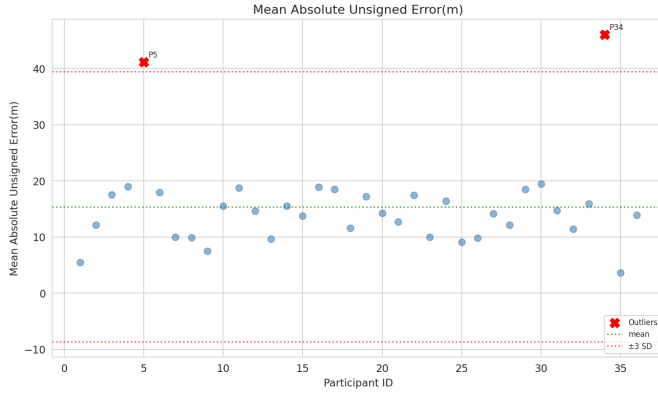


Figure 8: Outlier Detection for Distance Data

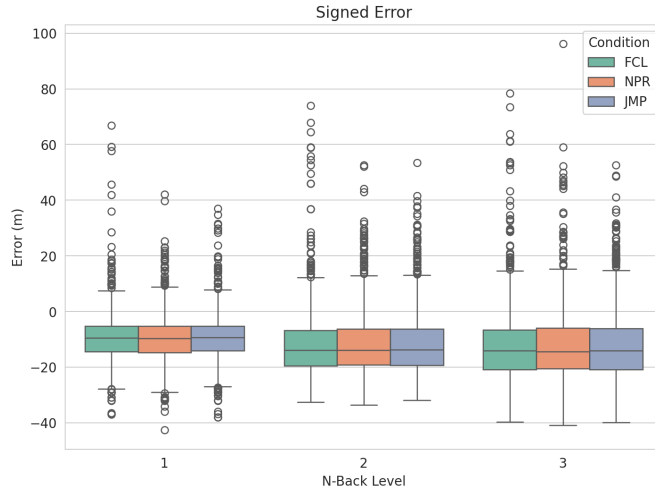


Figure 9: Distance Error After Removing Outliers

effect on signed distance error ( $\chi^2 = 1.12$ ,  $p = 0.572$ ). All three sessions caused similar underestimation biases (Session 1:  $M = -10.68$ m; Session 2:  $M = -10.66$ m; Session 3:  $M = -10.62$ m). This indicates that practice and fatigue did not influence how participants judged distance.

#### 4.2.3 Map Drawing

A rater scored sketch maps on three criteria: correct number of rooms (0-6), accurate door placement (0-6), and correct artifact descriptions (0-6), which sum to a total score of 0-18. A Friedman test showed no significant effect of locomotion method on total error in map drawing ( $N = 36$ ,  $\chi^2 = 4.24$ ,  $p = 0.120$ ); FCL ( $M = 2.11$ ,  $SD = 3.12$ ), JMP ( $M = 2.19$ ,  $SD = 2.21$ ), and NPR ( $M = 1.81$ ,  $SD = 3.45$ ). Floor plans also showed no significant effect on total error ( $\chi^2 = 2.62$ ,  $p = 0.270$ ), which confirms similar design across the three virtual environments.

| View | Total Error |      |        |     |      |
|------|-------------|------|--------|-----|------|
|      | Mean        | SD   | Median | Min | Max  |
| FCL  | 2.11        | 3.12 | 0.5    | 0.0 | 11.0 |
| JMP  | 2.19        | 2.21 | 2.0    | 0.0 | 7.0  |
| NPR  | 1.81        | 3.45 | 0.0    | 0.0 | 16.0 |

Table 2: Map drawing statistics for total error by locomotion method.

#### 4.2.4 Object Recognition

Participants viewed 12 images (6 present in their environment, 6 absent) and answered whether they recalled seeing each artifact. Overall recognition accuracy was high across all conditions, with no significant differences by Friedman test ( $\chi^2 = 0.42$ ,  $p = 0.809$ ). Mean scores and variability across conditions were comparable as shown in Table 3. This result suggests that object encoding occurred mainly during stationary viewing periods at each waypoint. The false positive rate ( $M = 35.0\%$ ,  $SD = 22.3\%$ ) was slightly higher than the false negative rate ( $M = 30.4\%$ ,  $SD = 21.7\%$ ) in all floor plans. A Wilcoxon signed-rank test indicated no significant difference between the two error types ( $W = 267$ ,  $p = 0.299$ ).

| Condition | N  | Mean | SD   | Median | Min | Max  |
|-----------|----|------|------|--------|-----|------|
| FCL       | 36 | 7.78 | 2.07 | 8.0    | 3.0 | 11.0 |
| JMP       | 36 | 8.03 | 1.98 | 8.0    | 5.0 | 12.0 |
| NPR       | 36 | 7.75 | 1.90 | 8.0    | 4.0 | 11.0 |

Table 3: Object Recognition statistics by condition.

#### 4.2.5 Per-Session Experience Ratings

After each VR tour, the participants rated their VR experience in four different aspects (1 = not at all, 4 = very much) including: enjoyment ("How much did you enjoy using this travel method?"), presence ("To what extent did you feel immersed in the virtual museum, as if you were really there?"), naturalness ("How natural did your experience feel overall?"), and realism ("How real did the virtual museum feel to you?"). Friedman tests showed no significant effect of locomotion method on any aspect: enjoyment ( $\chi^2 = 1.83$ ,  $p = 0.400$ ), presence ( $\chi^2 = 4.74$ ,  $p = 0.094$ ), naturalness ( $\chi^2 = 5.57$ ,  $p = 0.062$ ), and realism ( $\chi^2 = 0.18$ ,  $p = 0.912$ ). FCL rated highest on presence  $M = 2.14$  and naturalness  $M = 1.64$ , while JMP got the lowest in presence  $M = 1.72$  and naturalness  $M = 1.31$ . This suggests that continuous locomotion supports stronger presence in the VR. JMP received the highest enjoyment rating ( $M = 2.00$ ), while FCL and NPR got  $M = 1.56$  and  $M = 1.53$ . This suggests participants appreciated the comfort and efficiency of JMP although it brought lower immersion. Realism ratings were nearly the same across conditions (FCL:  $M = 1.89$ ; NPR:  $M = 1.83$ ; JMP:  $M = 1.81$ ).

#### 4.2.6 Exit Survey: Overall Preference and Qualitative Feedback

In the last session, after they had experienced all locomotion methods, participants completed an exit survey in which they rated each locomotion method on a 0-10 scale. They also answered: "Which of the three travel methods did you prefer *most*, and why?" and "Which did you prefer *least*, and why?" A Friedman test on the 0-10 preference scores showed no significant difference across methods ( $\chi^2 = 0.35$ ,  $p = 0.838$ ; Table 4). 15 participants (41.7%) named JMP as the method they preferred most, 9 (25.0%) named FCL, and 8 (22.2%) named NPR. Interestingly, JMP was also the most frequently named as the least preferred method (16 participants, 44.4%), with NPR named by 9 participants (25.0%) and FCL by 7 (19.4%).

For each locomotion method, positive themes were drawn from the free responses of participants who named that method as their most preferred, and negative themes from those who named it as their least preferred. Responses were categorized with the assistance of Claude [3] and verified by the author.

1. FCL: Participants who preferred FCL valued its realism and sense of immersion. P9 explained, "I prefer the full color the most, because it gives me a sense of engagement, and it is very natural to me. Also, it gives me a sense of direction and where I am going." Similarly, P32 noted that with FCL, "You could better place yourself in the environment as you moved," while P17 appreciated that "we can remember everything, so it causes to not lost during journey." However, FCL was also the most frequent cause of cybersickness symptoms. P6 reported, "I prefer full color moving method least. It would bring a lot of dizziness

and cybersickness during the rotation after you just arrive at one standing point. What's more, it takes a long time to move between points." P26 described the cybersickness: "The full color slide/rotate method caused really bad sickness, especially when turning far." P21 simply stated "was a lot of sickness associated with it for me," and P30 found it "definitely by far caused me the most nausea," caused by "the extra information that was provided by the colors." Even supporters acknowledged the trade-off, like P9 saying "there is a tradeoff because I felt sick most in this method."

2. NPR: The NPR method was described as a compromise between immersion and comfort. P11 explained, "I prefer the Non-photorealistic Rendering with black and white filter the most. It offers me more sense of orientation but avoid giving me too much visual information to cause me motion sickness when moving and rotating." P16 elaborated that "NPR felt like a balance of the two, minimizing the motion sickness but also still feeling like I was traveling through the museum. Having just the outlines of the objects also had me guessing what it would look like and what colors it would be before the image appeared." P27 found that "it didn't strain my eyes as much, but it wasn't as disorientating as the teleportation one," while P10 simply appreciated "less dizziness and more fun." P22 noted that "Non-photorealistic decreases the non-necessary details but preserve the outline of objects, and it is more easily for me to memory the objects I saw and the spatial information of the room." However, critics found NPR distracting and artificial. P1 complained the method was "super bright and was heavily irritating. I didn't enjoy the travel method at all compared to the other two methods." P7 reported "eye strain from the rapid shift to brightness," and P3, an experienced VR user, found "the cuts to white with black outlines were distracting if anything." P34 disliked that "the change in color made it difficult to focus and made the experience less enjoyable," while P31 felt "the black and white and the transition to colored make me feel less real."
3. JMP: Participants described JMP's main advantage as its efficiency and comfort. P23 found "the teleportation was the most convenient of the three; it took the least amount of time," while P34 preferred it "because it was quicker and did not make me dizzy at all." P26 stated that "it doesn't induce any motion sickness at all. The other two methods caused sickness when you viewing direction doesn't match the movement direction perfectly." P30 appreciated that "it was instant, and there was no movement that could cause me to feel nauseous, especially the rotating in the other two methods." P6 valued the efficiency: "It takes less time for moving and rotation. I am not very patient when I wait in the transitions between different points." P28 found similarity with video games helpful: "the jump method because it was the least disorienting, and I have played games with that style before." However, some pointed out the teleportation's impact on spatial awareness. P9 stated, "I don't like the jump method because I totally lose sense of orientation and where I am at. It doesn't feel natural at all to me." P22 elaborated: "The jump method is my least favor one. It didn't give me any clue of the route I traveled, so I couldn't remember the spatial arrangement of the museum and wasn't able to remember where I started from." P16 found it "really disorienting how I was positioned relative to my previous location (rotation wise) and I kept needing to look around before and after to judge where I was. Judging distances from the previous location was harder because you had no temporal sense of how far you had traveled." P13 summarized the trade-off: "I forgot the orientations and the previous moves I made. Although it was the most comfortable and quickest movement." Even P6, who preferred JMP, acknowledged "the only disadvantage of jump is that your spatial awareness would be destroyed a little bit and you would get lost in direction more often."

| Condition | N  | Mean | SD   | SEM  | Median |
|-----------|----|------|------|------|--------|
| FCL       | 36 | 5.69 | 2.70 | 0.45 | 6.0    |
| NPR       | 36 | 5.17 | 2.38 | 0.39 | 5.0    |
| JMP       | 36 | 5.39 | 2.82 | 0.47 | 6.0    |

Table 4: Exit Survey Preference Scores by Condition

## 5 DISCUSSION

This research addresses a barrier to VR adoption in architectural design, which is lacking comfortable guided tours that support spatial understanding. Our findings demonstrate that applying NPR during continuous locomotion achieves a balance across multiple measures, which can be summarized against the hypotheses posed in Section 2.3. H1 was supported by the  $\Delta$ SSQ results: NPR produced significantly less post-exposure cybersickness than FCL, and the ordering of both FMS and  $\Delta$ SSQ means ( $FCL > NPR > JMP$ ) matched the predicted relationship between transitional visual motion and discomfort. H2 was supported: NPR produced significantly lower pointing errors than JMP. H3 was supported by the Bayesian equivalence analysis showing that NPR and FCL pointing accuracy were statistically equivalent. In the following, we move from these observed effects to the mechanisms that we believe explain them, and then to their broader implications.

### 5.1 Cybersickness reduction

$\Delta$ SSQ Total ( $p = 0.018$ , Holm-Bonferroni corrected) showed NPR reduced post-exposure cybersickness significantly compared to FCL. The reduction was most noticeable for nausea ( $\chi^2 = 14.28$ ,  $p < 0.001$ ).  $\Delta$ SSQ ( $p = 0.120$ ) did not show a significant difference between NPR and JMP.

The SSQ-based and FMS-based evidence should be distinguished, as the two instruments play complementary roles. FMS is an instantaneous self-report obtained at each waypoint and captures the evolution of cybersickness over time; the post-exposure SSQ is administered after the entire session and is akin to a more detailed interpretation of the last-obtained FMS report, decomposed into symptom subscales. Cybersickness takes time to evolve: in the beginning, all three conditions were tolerable, and the full-session pairwise FMS differences were accordingly not statistically significant. The differences started to become apparent only after sustained exposure, with FCL producing significantly higher FMS than JMP, and NPR than JMP, from station 11 onward.  $\Delta$ SSQ reflects the total increase in cybersickness between the beginning and end of the session, whereas FMS captures the trajectory leading to it. The strongest claim our data support is therefore the SSQ-based one, that NPR significantly reduced accumulated post-exposure cybersickness compared to FCL, while the FMS data show that this difference emerged progressively over the session rather than being present from the outset.

An important finding is the correlation between VIMSSQ and  $\Delta$ SSQ, where self-reported motion sickness susceptibility predicted cybersickness in JMP ( $r = .363$ ,  $p = .030$ ) and FCL ( $r = .314$ ,  $p = .062$ ), but not in NPR ( $r = .082$ ,  $p = .636$ ). This finding suggests that sketch-style rendering may reduce cybersickness symptoms by reducing the high-frequency visual detail that increases symptoms in susceptible individuals. A broader implication is that NPR may act as an equalizer: rather than merely lowering average discomfort, it appears to decouple individual vulnerability from symptom outcomes, which is particularly valuable for client presentations where the audience's susceptibility is unknown in advance.

### 5.2 Spatial awareness preservation

NPR locomotion preserved spatial orientation comparable to FCL and significantly better than JMP. The Friedman test showed a significant locomotion effect on unsigned pointing error ( $\chi^2 = 15.76$ ,  $p < 0.001$ ,  $N = 34$ ), with FCL ( $M = 10.60^\circ$ ) and NPR ( $M = 11.59^\circ$ ) not differing significantly ( $p = 0.143$ ). Both FCL and NPR outperformed JMP ( $M = 15.10^\circ$ , both  $p < 0.001$ ). A Bayesian equivalence analysis provided strong evidence that NPR and FCL pointing accuracy were statistically equivalent ( $BF_{01} > 1000$  across all tested equivalence margins and prior

scales). Baseline spatial orientation test error was strongly correlated with angular pointing error ( $r = 0.699$ ,  $p < .001$ ), which confirms that spatial ability contributes to spatial updating task performance in VR. An ANCOVA test confirmed that SOT was a significant covariate ( $F(1, 102) = 53.98$ ,  $p < .001$ ) and showed a significant Locomotion method  $\times$  SOT interaction ( $F(2, 102) = 5.17$ ,  $p = .007$ ). This result suggests that pointing accuracy depends more on spatial ability when visual feedback is reduced or absent during transition.

Why did the locomotion method affect pointing accuracy but not distance estimation, map drawing, or object recognition? We believe these measures tap different underlying processes. The pointing task assesses spatial updating, the continuous maintenance of self-position and orientation during travel, which is precisely the process that benefits from the continuous transitional visual feedback provided by FCL and NPR. Klatzky et al. [26] found that physical rotation with visual feedback was sufficient for accurate spatial updating even without translational body-based cues, while purely visual movement led to the greatest errors. Related work indicates that it is passive rotation without idiothetic cues that reliably harms spatial updating in VR, whereas passive translation does not appear to carry the same cost [6, 37]. Since all movement in our experiment was passive, the critical moment for spatial updating may be the rotation upon arrival at each waypoint: during NPR and FCL this rotation is a continuous visual motion providing optic flow cues that support orientation tracking, whereas in JMP the rotation is instantaneous, eliminating those cues entirely. This account explains why pointing errors, which reflect rotational orientation knowledge, were significantly worse under JMP. In contrast, map drawing and object recognition may depend more heavily on information that can be as readily acquired during stationary viewing at each waypoint, which was identical across conditions; the null condition effects for these measures ( $\chi^2 = 4.24$ ,  $p = 0.120$  and  $\chi^2 = 0.42$ ,  $p = 0.809$ , respectively) are consistent with this interpretation. Distance estimation was likewise unaffected by locomotion method ( $\chi^2 = 1.65$ ,  $p = 0.439$ ), with all conditions producing similar underestimation biases (approximately  $-10.7\text{m}$ ), and underestimation increased with  $n$ -back level ( $\chi^2 = 26.12$ ,  $p < 0.001$ ). However, we note that the verbal distance judgments, unlike the physically produced pointing responses, were characterized by exceptionally large errors, and this measure may be a somewhat fraught metric in the context of our experimental design. Taken together, the pattern suggests that “spatial awareness” is not a unitary construct: transitional visual information selectively supports the online updating of self-orientation, while layout memory, content memory, and coarse distance representations are built primarily from stationary viewing, and therefore remain robust to the rendering style, or even the absence, of the transition.

### 5.3 User experience

User experience ratings collected after each session showed locomotion method had no significant effects on enjoyment, presence, naturalness, or realism. In the exit survey, where participants compared all three methods, preferences were diverse: 15 participants (41.7%) named JMP as their most preferred method, while 16 (44.4%) named it as their least preferred. FCL was most preferred by 9 (25.0%) and NPR by 8 (22.2%). NPR was rarely named as least preferred, which shows that our method was able to avoid the worst shortcomings of JMP and FCL. The qualitative responses support these findings. Participants described NPR as providing “a balance”, as it provides enough visual information to maintain orientation while avoiding “too much visual information to cause me motion sickness.” Some users appreciated “guessing what it would look like” before full detail appeared, which can be a cognitive engagement similar to how architects use progressive detail revelation during design presentations.

## 6 LIMITATIONS AND FUTURE WORK

The participant sample was drawn from a university population and may not fully represent the population of architectural clients. Because our results suggest that NPR stylization specifically benefits individuals with higher motion sickness susceptibility, the practical value of NPR

locomotion may extend to, or even grow in, populations whose susceptibility profile differs from our sample’s; validating this with practicing architects and their clients is a natural next step.

Our virtual museums differ from real architectural designs in level of detail and production value. While our models’ unpredictable layout and high texture detail afforded good control over our experimental variables, the models we used do not capture the diversity of residential, office, or mixed-use designs, or the production quality of walkthrough-ready architectural models. Additional testing with such models, ideally in collaboration with practicing designers, would strengthen the external validity of our results.

Our study examined passive, guided locomotion along predefined paths. We deliberately restricted the scope in this way because passive movement is consistent with the guided-tour concept that motivated the work. Consequently, we are reluctant to extrapolate our findings to other contexts. Active locomotion, open-ended exploration, training simulations, and VR games each impose different attentional and navigational demands, and provide different affordances for cybersickness compensation interventions. Whether the comfort–cognition benefits of NPR transitions can transfer other settings is an open empirical question that our paradigm can be adapted to test.

Finally, in our NPR rendering we only explored a single sketch-like aesthetic based on depth and surface-normal edges. While this approach worked well in our simple indoor environments, it may fall short in settings with dense organic geometry such as outdoor scenes with foliage. Our account attributes the comfort benefit to the removal of high-frequency detail from the transitional optic flow; if that account is correct, alternative stylizations (e.g., watercolor, blurring) that differ in how much structure they preserve should produce different comfort–cognition trade-offs. Selective stylization of only the visual periphery, where optic flow is strongest, might retain the comfort benefit while further improving the user experience. Evaluating these variants against our account is a promising direction for future work.

## 7 CONCLUSION

VR adoption in architectural design depends on resolving the trade-off between affording comfortable movement and supporting spatial understanding. Our research demonstrates that deploying NPR stylization during architectural guided tour transitions can achieve this resolution. Post-exposure cybersickness was significantly reduced in our NPR condition compared to FCL ( $\Delta\text{SSQ Total}$ :  $p = 0.018$ ), while NPR and JMP showed no significant difference ( $p = 0.120$ ).

NPR stylization supported spatial awareness statistically equivalently as well as FCL. Pointing accuracy differed by approximately  $1^\circ$  (Bayesian equivalence  $\text{BF}_{01} > 1000$ ), and performance was significantly better in the NPR and FCL conditions compared to JMP ( $p < 0.001$ ). Distance estimation, map drawing, and object recognition were unaffected by locomotion method, suggesting that these spatial cognition measures may depend more heavily on information that can be derived from stationary views. As discussed in Section 5, NPR stylization preserves the continuous rotational visual cues that support spatial updating while reducing the high-frequency textural detail in the optic flow that drives cybersickness [13, 40]. Our findings offer a balanced solution for guided virtual architectural tours that were previously constrained by cybersickness. Using NPR stylization to mitigate cybersickness during transitions enables the use of continuous virtual locomotion to help clients develop a comprehensive spatial understanding of designs and make more informed decisions.

## ACKNOWLEDGMENTS

The experimental framework was drawn from K. Vaziri’s [45] thesis work on NPR in architectural VR. Museum assets were obtained from Unity Asset Store vendors. The authors used Claude [3] to assist with grammar checking and language editing of this manuscript.

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